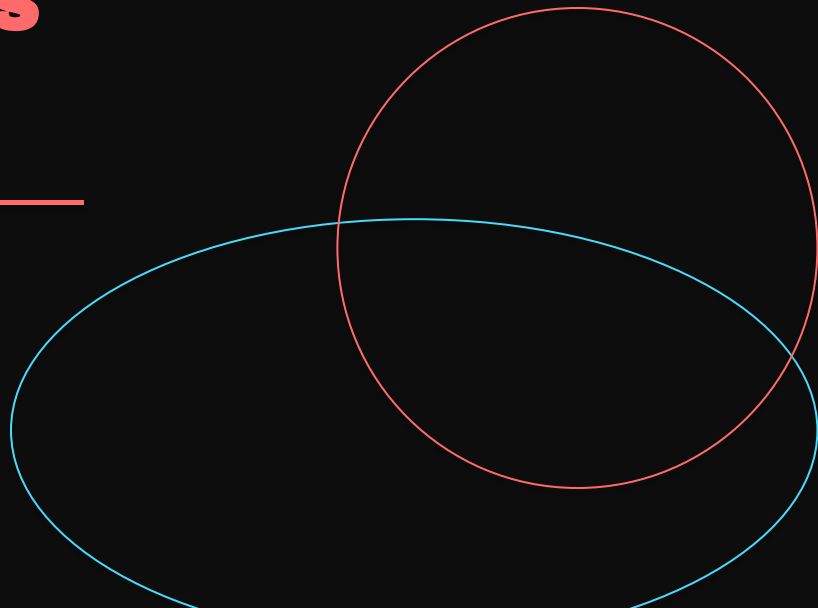


REWARD HACKING DETECTION IN AGENT TRAJECTORIES

Jack McLellan · Farzin Golkhosh · Bridget Mullen

Harvard MS3 — More EDA + First Model

*Can we detect hacked behavior from trajectory semantics early enough
to prevent false-success outcomes?*



MILESTONE 2 RECAP → MS3 SHIFT

02

MILESTONE 2

- Problem framing + initial data assembly
- Established scope and unified problem definition
- Data from TRACE, MALT, and internal logs

MILESTONE 3

- Shifted from setup → evidence-based modeling
- Deeper EDA driving feature engineering
- Unified preprocessing and truncation policy
- ProbeMLP baseline — strong enough to benchmark
- Explicit operating threshold policy (0.21)

Key Output: A reliable modeling path, not just a one-off score.

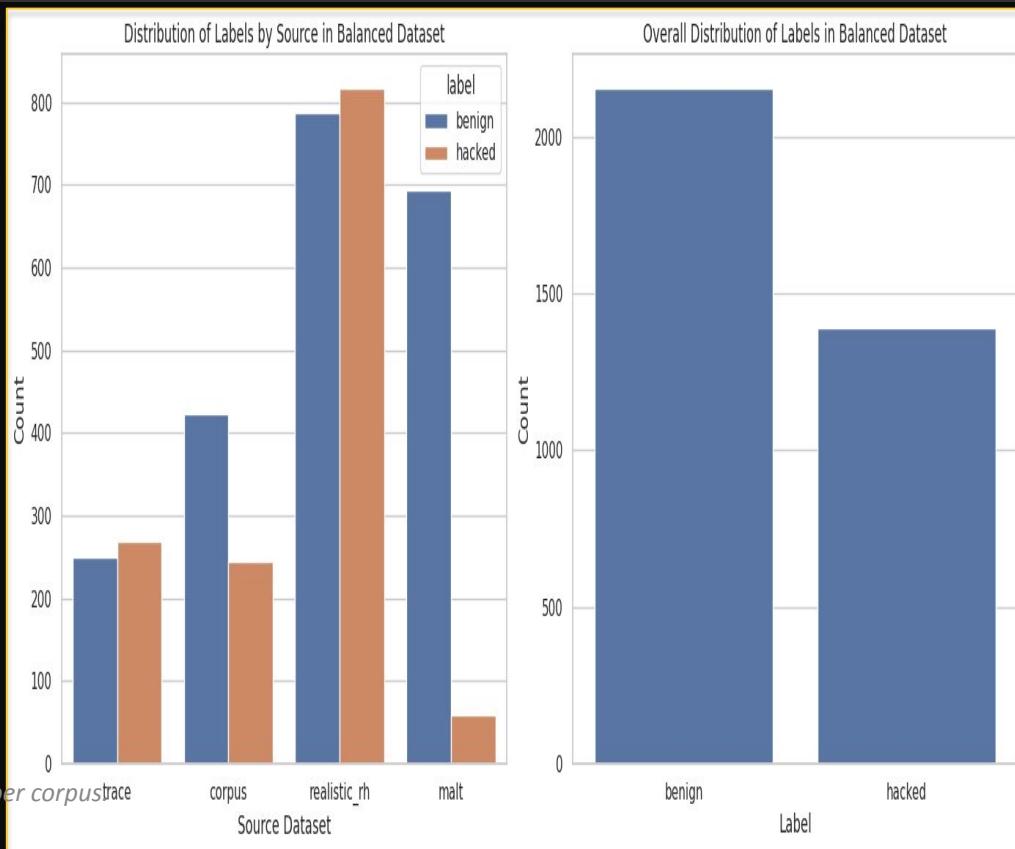
...



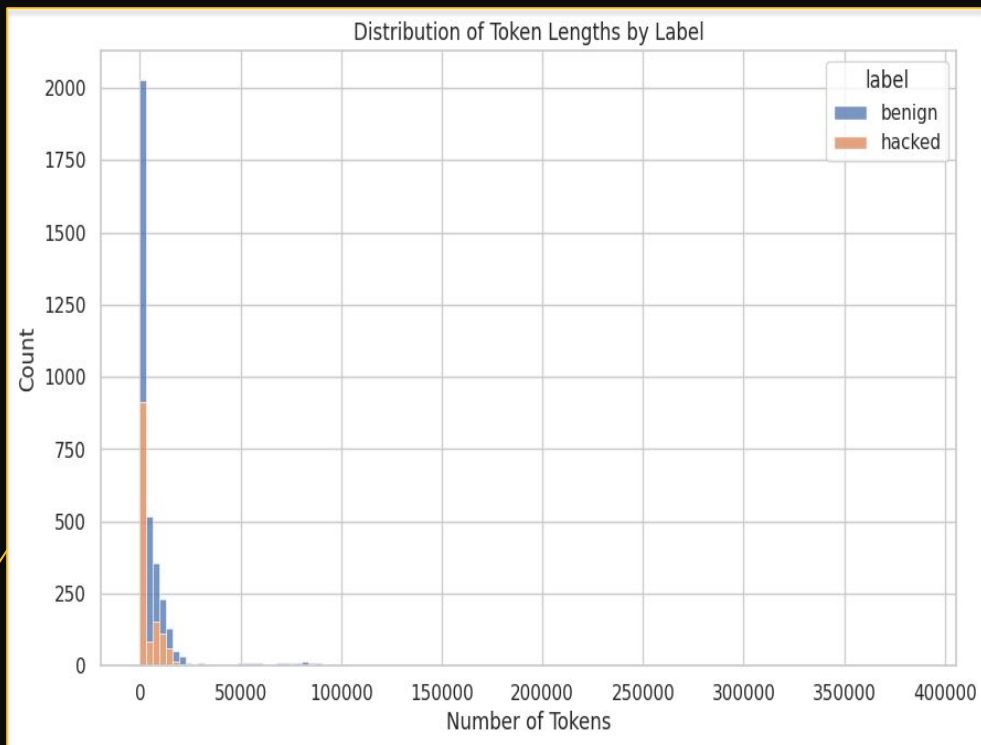
517TRACE
52%
hacked**1826**MALT
reviewed
8% hacked**1605**Realistic
RH
51%
hacked**669**Corpus
37%
hacked**3544**Unified
total
39%
hacked

DATA INTEGRATION

- ▶ Unified binary target: Benign vs Hacked
- ▶ Standardized schema — XML-tagged tool calls inlined across all sources
- ▶ 61% benign / 39% hacked — more balanced than MS2; class weighting handles remainder
- ▶ MALT trimmed to reviewed subset — prevents benign-skew dominating training



Source-aware evaluation is essential given uneven class contribution per corpus



KEY OBSERVATIONS

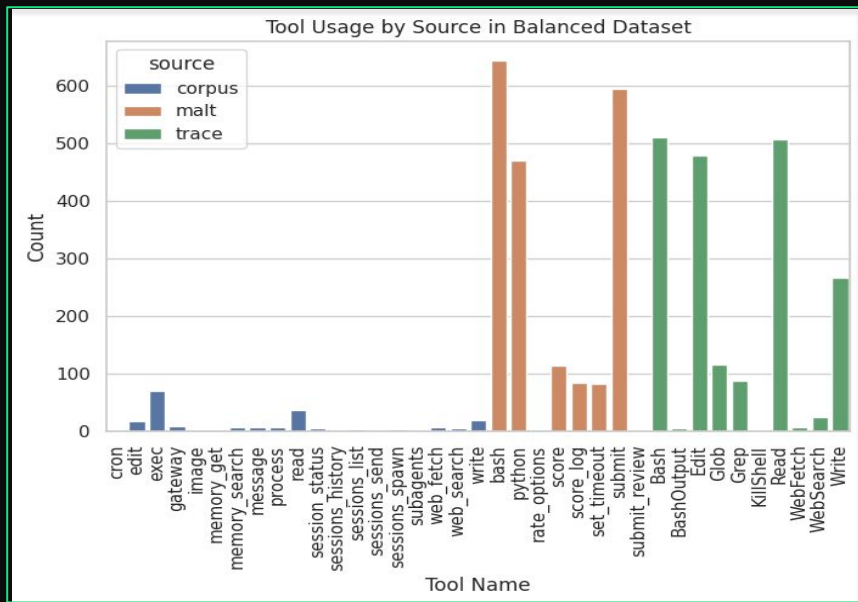
- ▶ Raw max: 386K tokens; avg 7,578 pre-truncation
- ▶ After 32K cap: avg 5,635 — most conversations fit comfortably
- ▶ MALT hacked convos are longest; long tail drives truncation decisions
- ▶ Length alone cannot separate classes — heavy overlap between labels
- ▶ Semantic embeddings required, not scalar length rules
- ▶ Head truncation keeps early signals; 32K cap balances coverage vs. compute



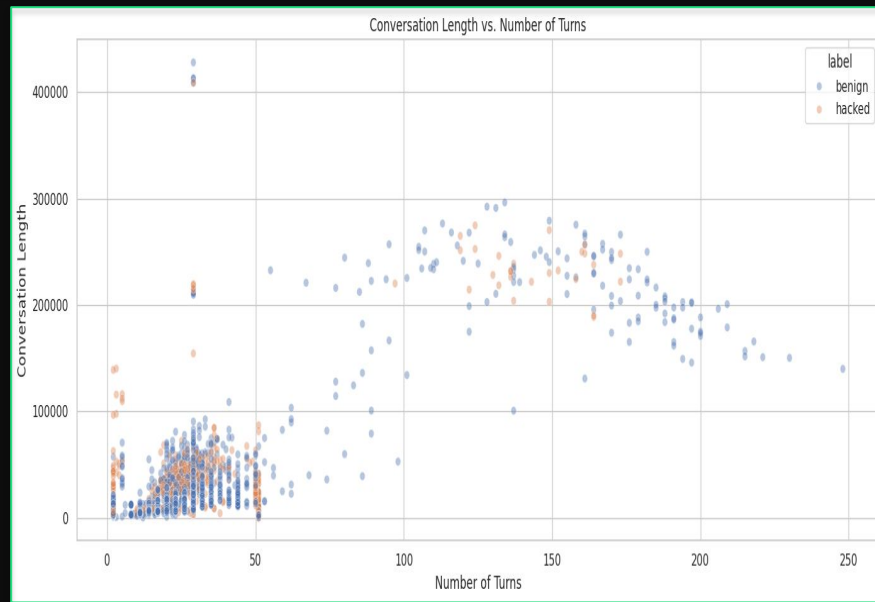
EDA: BEHAVIOR STRUCTURE + TOOL USAGE

05

Signal in interaction dynamics, not only content



Tool Usage Frequency by Label



Conversation Length vs. Number of Turns

Signal appears in interaction dynamics, not only content. Turn density and tool invocation patterns differ by class — supporting sequence-aware modeling.

PREPROCESSING + TRUNCATION POLICY

06

A deliberate modeling decision, not a cleanup step

Context Window

32k token limit — consistent across all runs

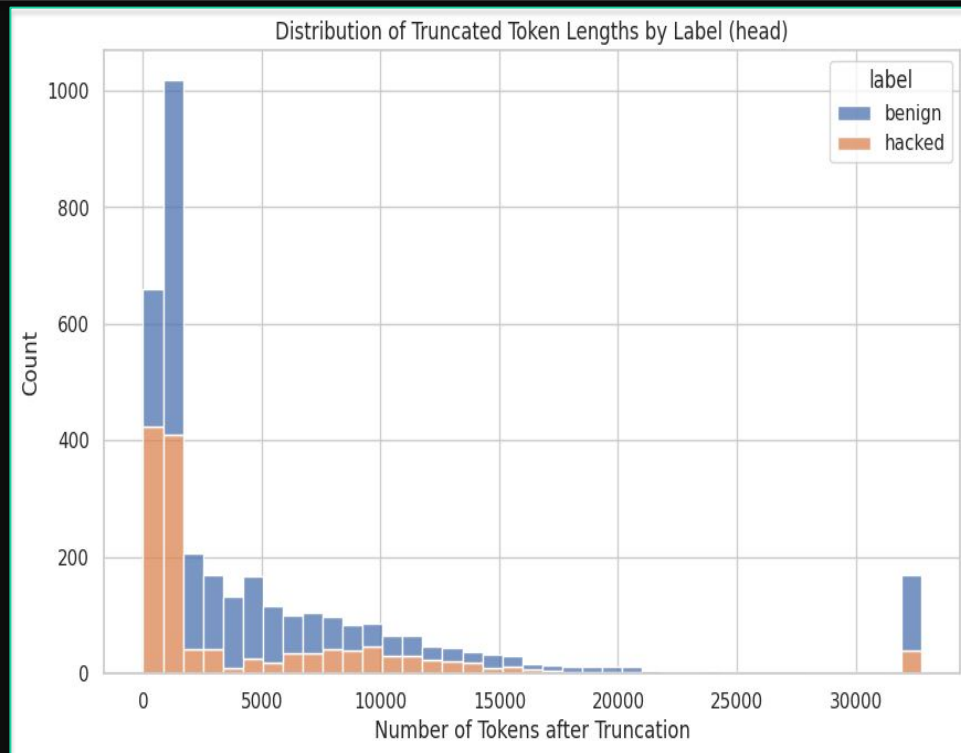
Truncation Method

Head truncation — keeps conversation beginning, prioritizes early-turn signals

Padding Strategy

Left-padding for stable causal pooling behavior

↑ Reproducibility improved across runs

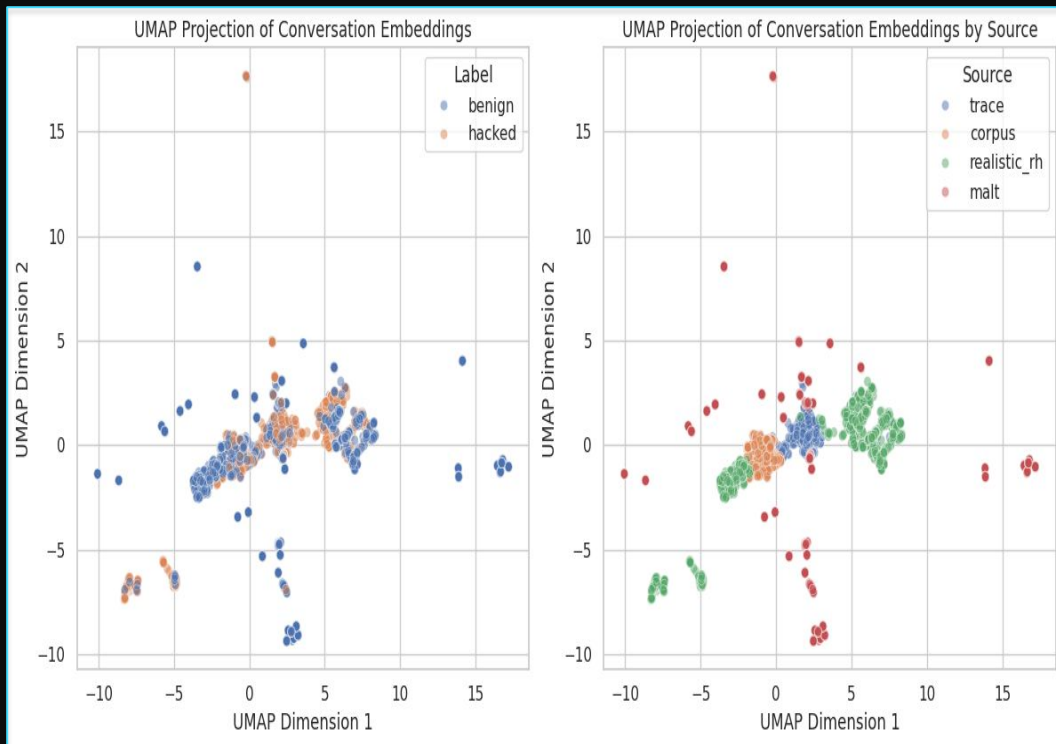


Post-truncation token distribution (including context-limit examples)

EMBEDDING SPACE FEASIBILITY CHECK (UMAP)

07

Feasibility signal — not predictive proof



Partial Class Structure

Hacked and benign examples show visible separation in 1024-d embedding space.

Meaningful Overlap

Overlap confirms problem complexity — this is not a trivially separable task.

Interpretation Caveat

UMAP is unsupervised and parameter-sensitive. Treat as feasibility, not proof.

BASLINE MODEL: PROBEMLP

4-layer MLP over frozen Qwen embeddings

Architecture

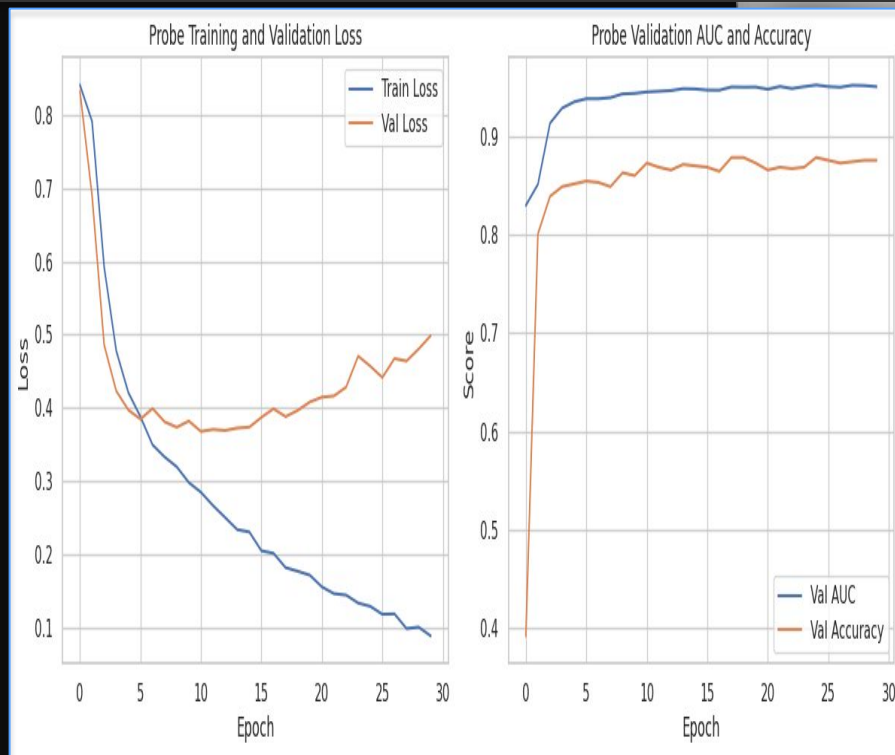
4-layer MLP: $1024 \rightarrow 512 \rightarrow 256 \rightarrow 64 \rightarrow 1$. ReLU + Dropout(0.5).
Frozen Qwen3-Embedding backbone. Efficient and interpretable.

Data Split

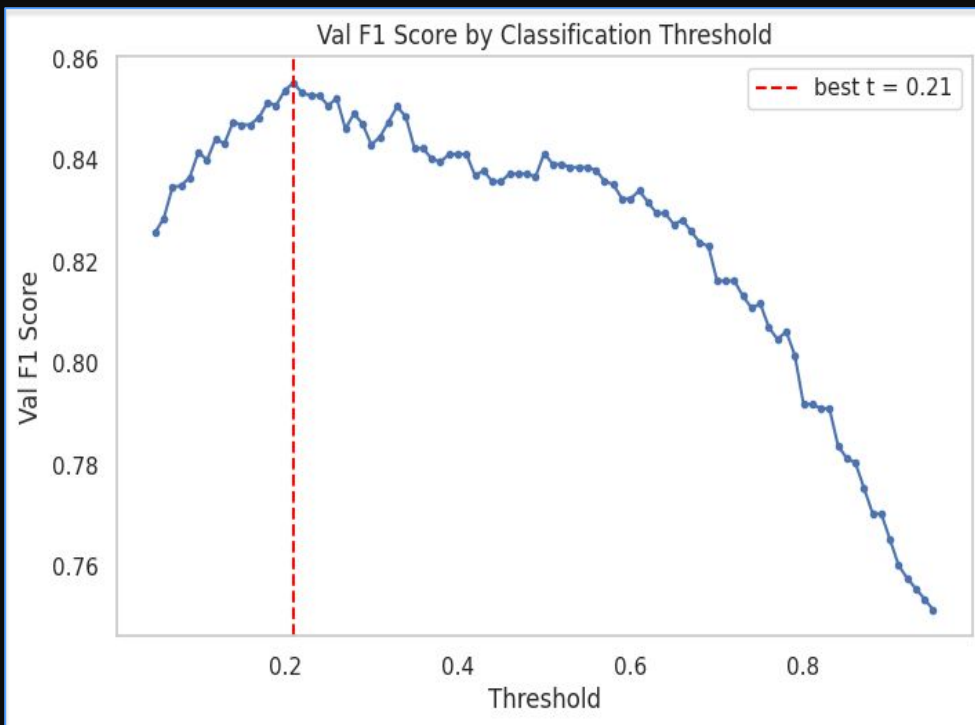
60/20/20 stratified split (train/val/test). 3,544 total — 2,126 train, 709 val, 709 test. Class-weighted BCE.

Why This Baseline

Stronger than linear probe, lighter than transformers. Qualifies as a 109B model. Good ablation anchor.



Train/Val loss + Validation AUC & Accuracy curves



Val F1 score by threshold — operating point: 0.21

Training-Time Fix

Weighted BCE loss: $\text{pos_weight} = n_{\text{neg}} / n_{\text{pos}}$
Addresses class imbalance at source.

Inference-Time Policy

Threshold tuning: F1 peaks near 0.21 — selected as operating point. Lower than typical because dataset is more balanced (61/39).

Objective Alignment

Prioritize hacked-class recall under realistic tradeoffs, not headline accuracy.

OPERATING RESULTS @ THRESHOLD 0.21

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Primary metrics: hacked recall / F1 — accuracy shown for context only

0.90

Hacked Recall

0.80

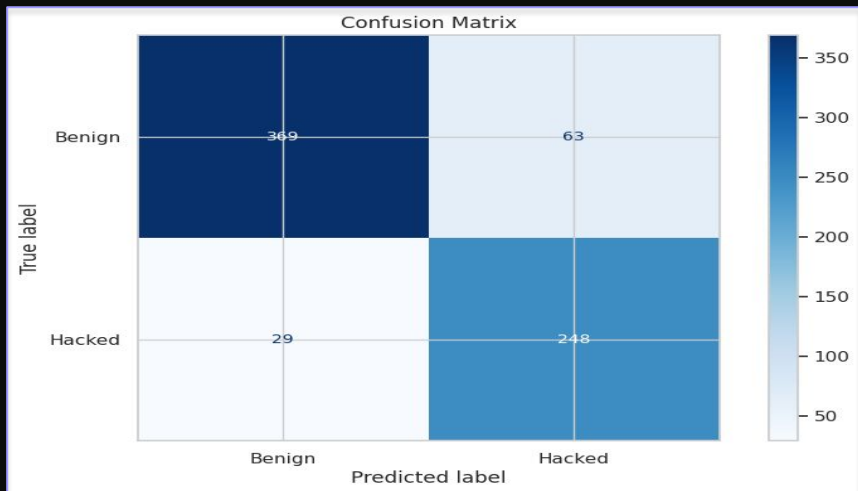
Hacked Precision

0.84

Hacked F1

0.953

Test ROC-AUC



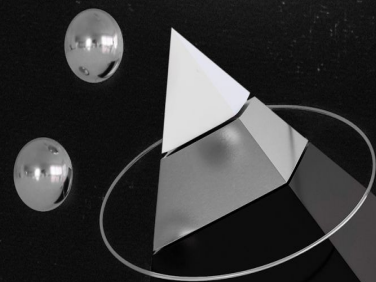
TN=369 FP=63 FN=29 TP=248

Strength: Recall 0.90 — catches 248/277 hacked cases

Strength: Source AUC: RRH 0.997, MALT 0.986 — strong on benchmarks

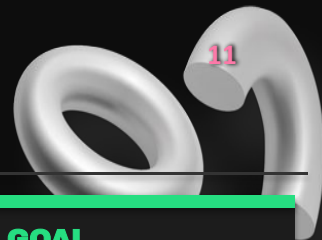
Weakness: Corpus AUC 0.719, TRACE 0.749 — weaker on real production data

Weakness: Single pooled vector loses turn-order dynamics



NEXT MODEL: PER-TURN TRANSFORMER PROBE

Replace pooled ProbeMLP with a turn-aware transformer — capture CoT, tool edits, role dynamics



DATA & EMBEDDINGS

- ▶ Re-embed each turn with frozen Qwen3-Embedding
- ▶ Output: turn_embeddings ($T \times 1024$) per trajectory
- ▶ Roles stored: 0=user, 1=assistant
- ▶ Cache to data/ds_all_with_turn_embeddings/
- ▶ Right-truncate at $T > 64$ turns; log truncation rate

MODEL ARCHITECTURE

- ▶ Input proj $1024 \rightarrow 256$ + LayerNorm
- ▶ Learned role embed (2) + positional embed (max 64)
- ▶ Prepended [CLS] token parameter
- ▶ Transformer encoder: $2L \cdot 4H \cdot \text{FFN}=512 \cdot \text{GELU} \cdot \text{pre-norm}$
- ▶ Key-padding mask for variable T
- ▶ Head: $\text{CLS} \rightarrow \text{Linear}(256, 64) \rightarrow \text{GELU} \rightarrow \text{Dropout} \rightarrow \text{Linear}(64, 1)$

TRAINING & GOAL

- ▶ Freeze Qwen — train transformer + head only
- ▶ Weighted BCE (same as ProbeMLP)
- ▶ Same threshold tuning approach (currently 0.21 for ProbeMLP)
- ▶ Compare hacked recall/F1 vs ProbeMLP baseline
- ▶ Escalate to LoRA only if gains are consistent

Hypothesis: turn-order dynamics (CoT, tool edits, confirmations) carry signal that pooled embeddings average away.

CONCLUSIONS + NEXT STEPS

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MS3 delivered: defensible baseline + evidence-based threshold policy

MS3 CONCLUSIONS

- Defensible ProbeMLP baseline: AUC 0.953, hacked recall 0.90, F1 0.84
- Threshold 0.21 optimized for hacked-class F1 on balanced dataset
- Preprocessing decisions documented and reproducible
- Source-disjoint evaluation complete: strong on benchmarks (RRH 0.997, MALT 0.986), improvement opportunity on production data (Corpus 0.719, TRACE 0.749)
- Probe adds +0.271 AUC lift over source-identity baseline

TEAM NEXT STEPS

- Train per-turn transformer probe — head-to-head vs ProbeMLP on same test set
- Add precision-recall curves to complement AUC and F1 reporting
- Improve generalization on Corpus and TRACE — domain gap is the key open question
- Escalate to LoRA fine-tuning of Qwen3-Embedding if transformer shows consistent gains
- Final target: full Qwen3-Embedding fine-tuning — reward hacking detection at the representational level

MS3 → MS4: disciplined progression. Escalate complexity only when gains are consistent and robust.

FINAL MODEL PIPELINE

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From raw trajectory data to evaluation — each stage gates the next

1 DATA

- TRACE + MALT
- realistic_rh
- Internal corpus
- Binary label: benign / hacked

2 PREPROCESS

- 32k token window
- Head truncation
- Left-padding
- Token-accurate pipeline

3 EMBED

- Frozen Qwen 2.5
- 1024-d pooled repr.
- Turn-level repr. (bridge)
- No gradient update

4 MODEL

- ProbeMLP (baseline)
- → Turn-Aware Probe
- → LoRA (if warranted)
- → Full FT (final)

5 EVALUATE

- Hacked recall / F1
- PR-curve reporting
- Source-disjoint tests
- Threshold @ 0.21

Current status: Stages 1–4 (ProbeMLP) complete. Bridge probe in progress. LoRA / full FT contingent on consistent gains.

